PHYC10009 Physics Foundations

Radiation Imaging
Week 11 Lecture 3

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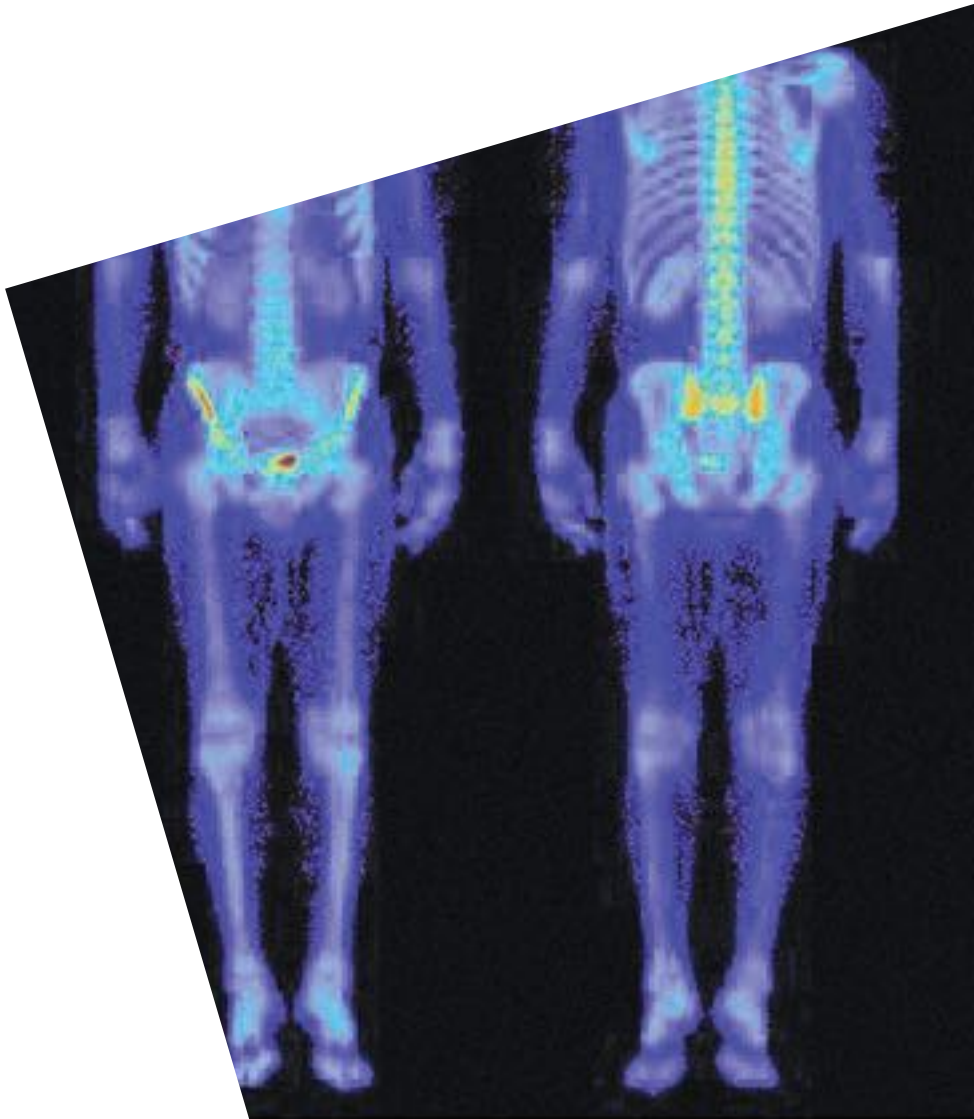
University of Melbourne



Todays lecture

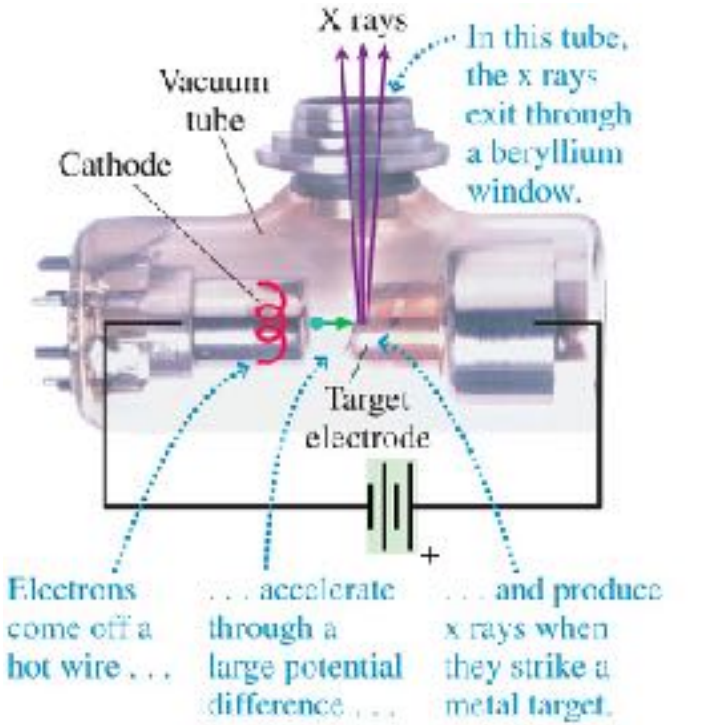
- Non-invasive imaging
- X-rays
- CT-scans
- Contrast X-rays
- Nuclear medicine
- PET scans
- Ultrasound imaging

X-rays

X-rays are a form of ionising radiation and provide the standard non-invasive diagnostic tool for medicine.

X-rays are photons in the energy range 5 KeV to 100 KeV.

They are normally generated by accelerating electrons through a high Voltage (up to 100 KeV) then colliding them with a metal target.



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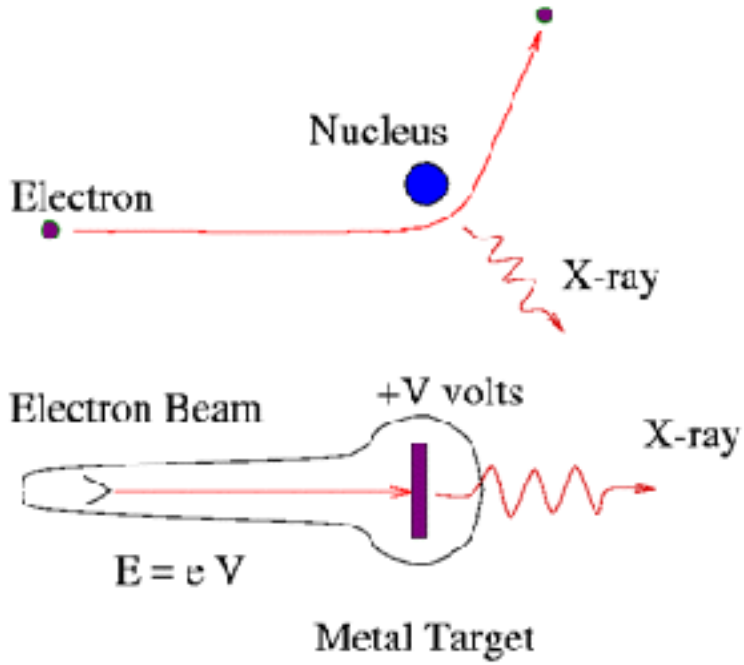
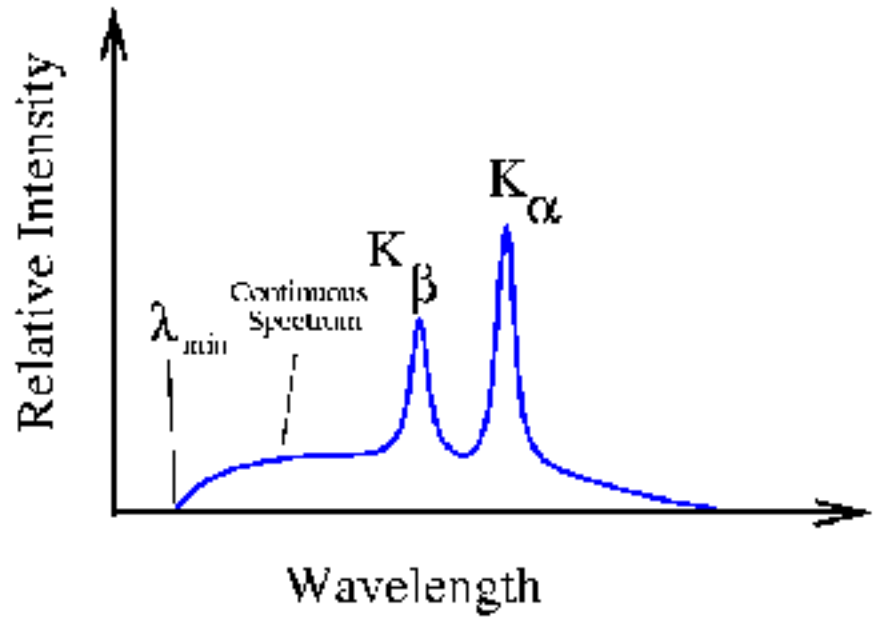
X-rays

This generates a spectrum of X-rays of energy up to that of the maximum Electron energy.





The peaks in the spectrum are due to collisions with inner-shell electrons
And the emission of high-energy photons as these electrons are replaced.

$$E_{photon}(max) = eV = hf_{max}$$

$\lambda_{min} = c/f_{max}$ $\quad$ $f_{max} = E_{elec}/h$ => $\quad$ $\lambda_{min} = hc/E_{elec}$

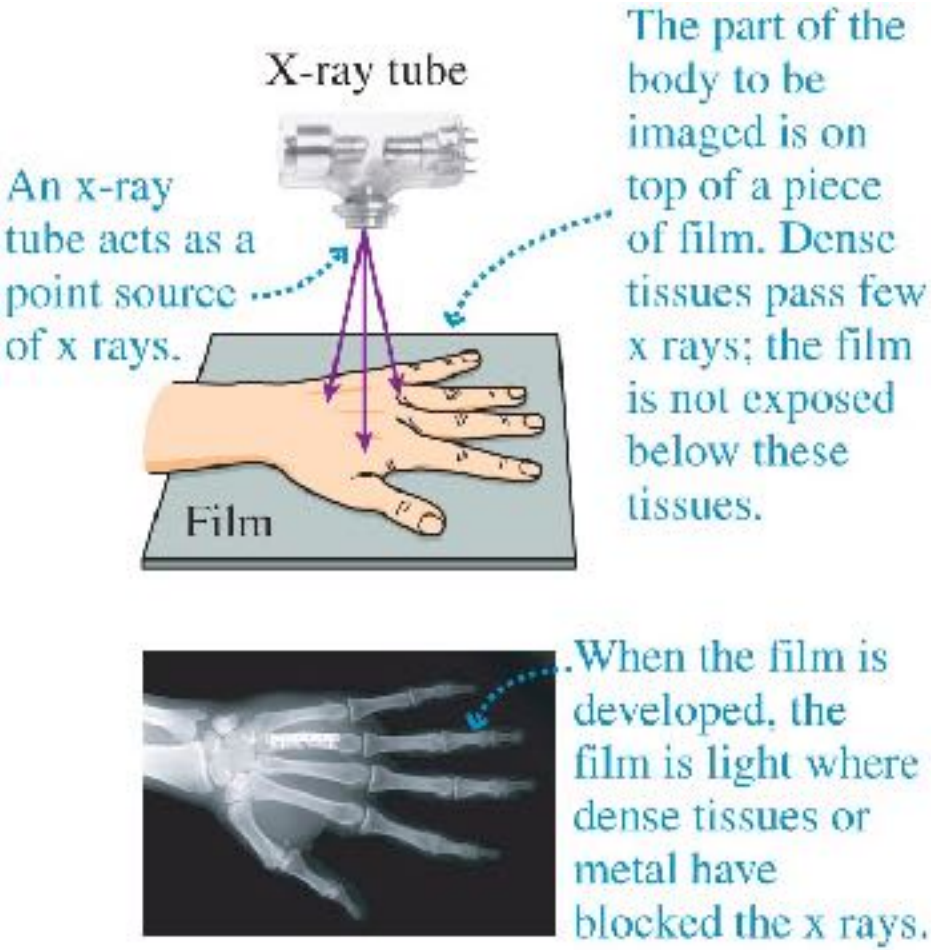
X-rays

When directed through the human body, X-rays are preferentially Absorbed by the Calcium in bones, which show up as shadows in the detection elements.

X-rays are useful as an imaging tool because they can penetrate bodily tissue allowing us to “see” inside the body.

However their use should be limited to minimise the cumulative radiation dose to Patients.

X-ray hand, arm = 0.01 mSv
X-ray chest = 0.06 mSv
X-ray pelvis = 0.65 mSv



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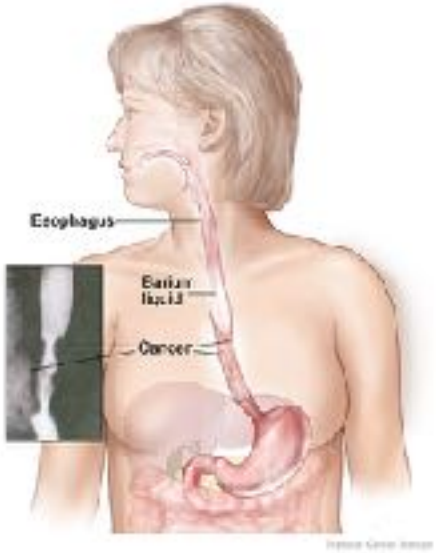
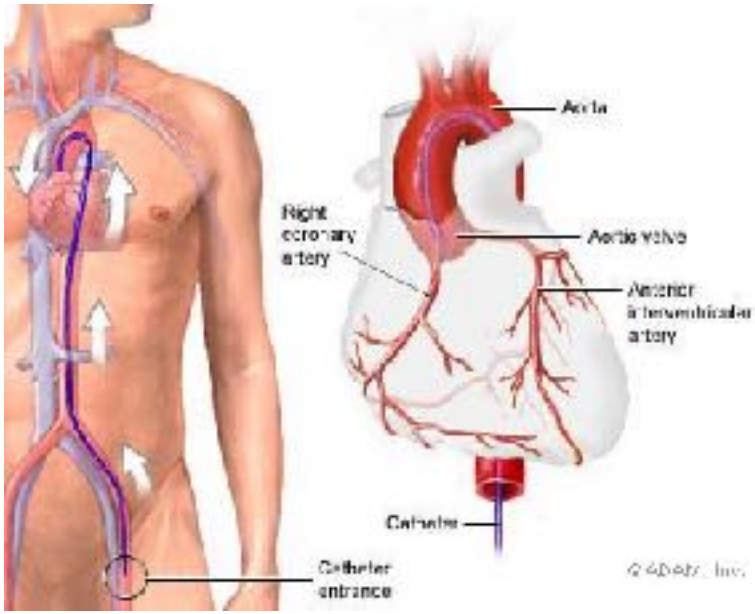
Contrast X-rays

The probability of X-ray absorption increase as Z^2 where Z is the atomic number of the material

So for example Barium with Z=56 is $\left(\frac{56}{6}\right)^2 =$ 84 times more effective at x-ray absorption than Carbon or Water

Internal bodily structure can be more easily determined through the use of a contrast agent together with X-rays.

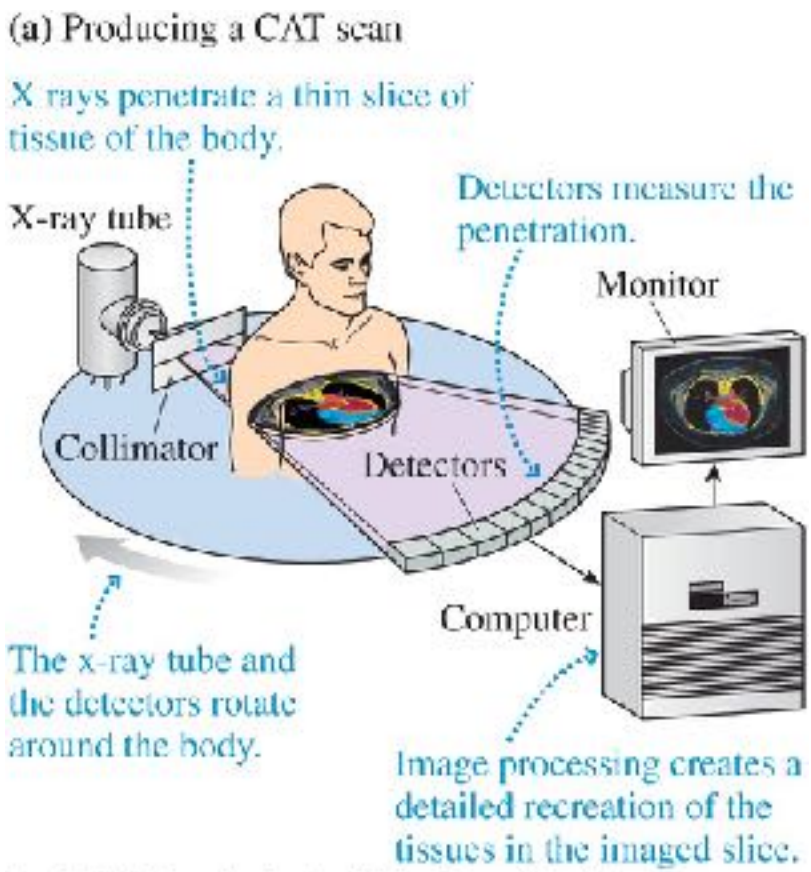




Computerised X-ray Tomography (CT-scans)

Tomography is the process of determining 3-dimensional structure by making many views of slices through the body.

This is done by rotating a gantry of X-ray source and receivers around the Body and recording the image at each angle.



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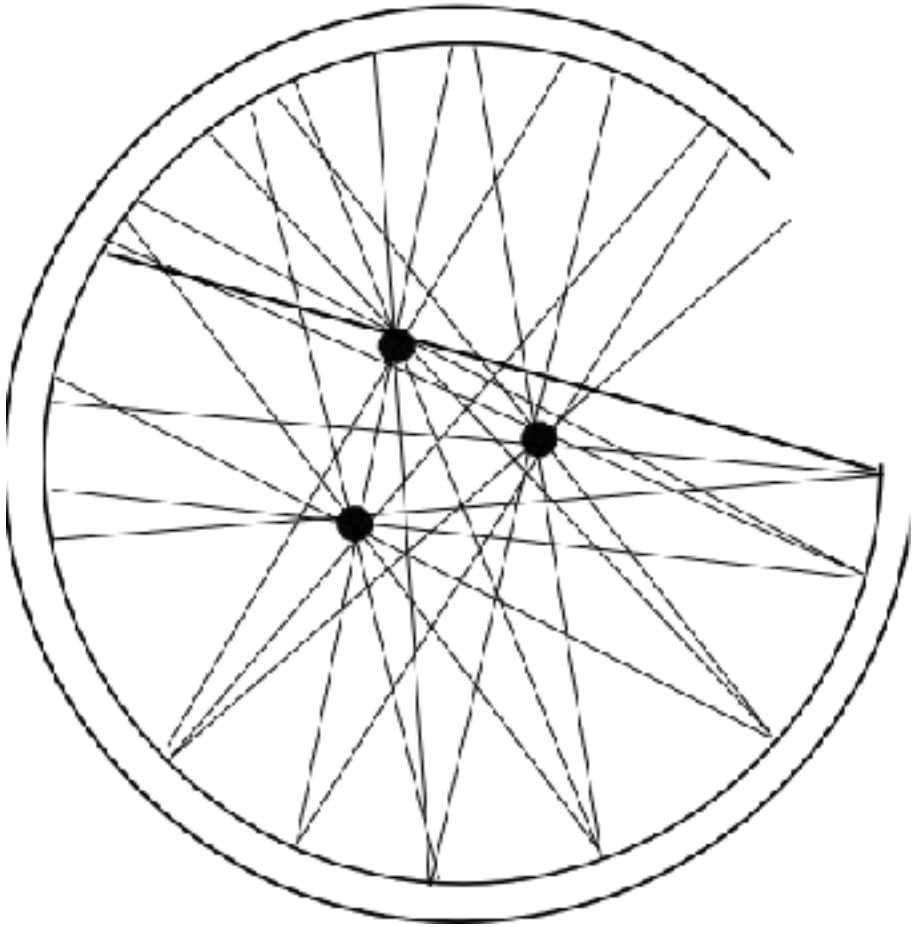
CT-scans

The position of the source together with the detector response is recorded over a wide range of angles.

These define numerous “lines of response” through body.



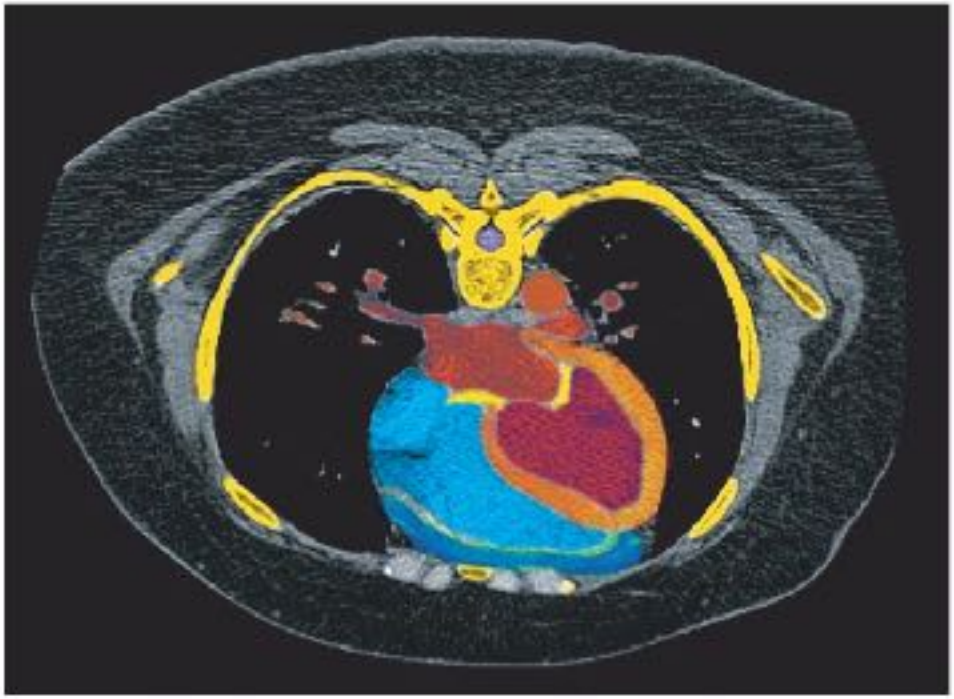
The intersection of these lines Show the locations of increased density

A computer is used to solve the numerous equations required to determine these Intersections and to reconstruct the image.

CT-scans

These scans are made at a variety of slices through the body which enables a full 3-D view of the internal structure.

(b) A CAT scan of the chest



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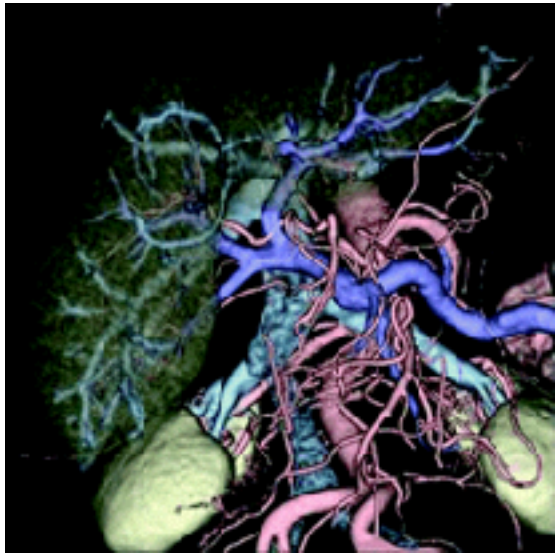
Contrast CT-scans

CT-scans can also be used to image soft tissue and other structures through the use of **High-Z** contrast agents



Contrast CT-scans

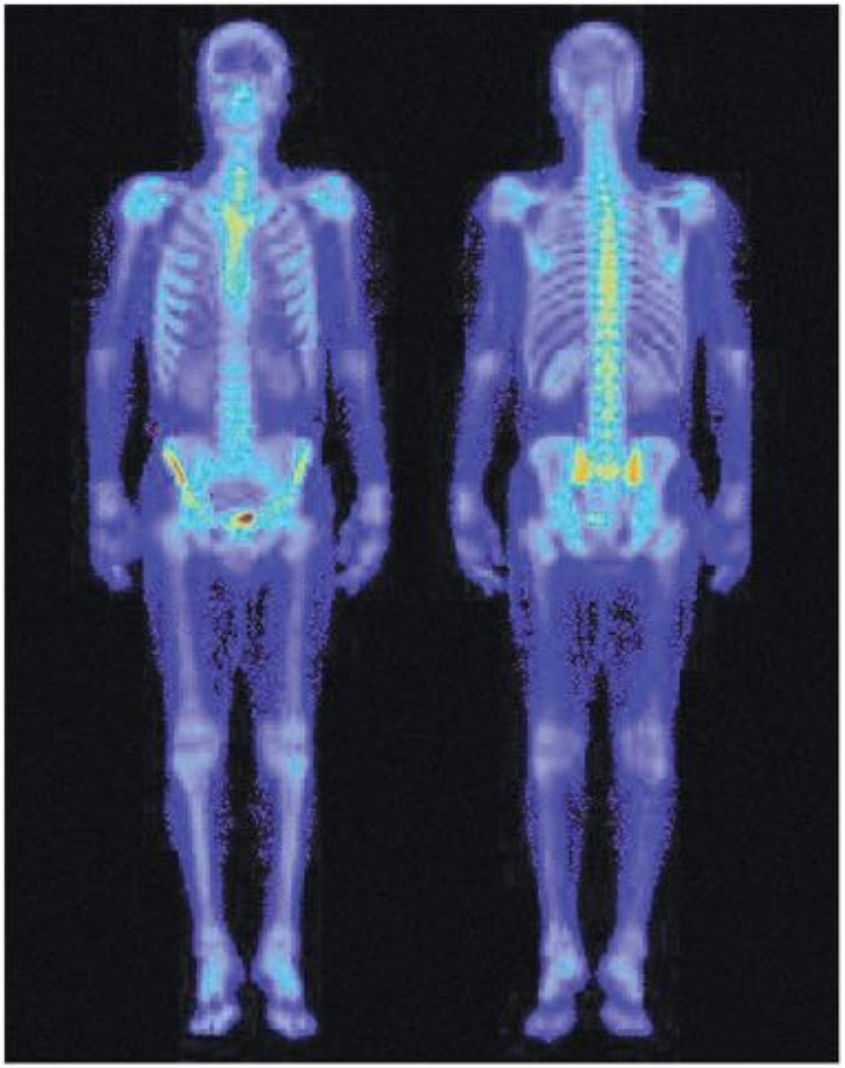
Nuclear Medicine

Whereas X-rays make images by directing radiation external to the body, Nuclear imaging employs radiation sources internal to the body.

The advantage of these techniques is that useful chemicals can be tagged by placing a radioactive Nucleus in the chemical.

The location of the chemical Is traced by locating where the Radiation originates from.

In the example shown the Patient is given a sample of ^{99}Tc which is taken up by bones in regions where active growth is occurring.



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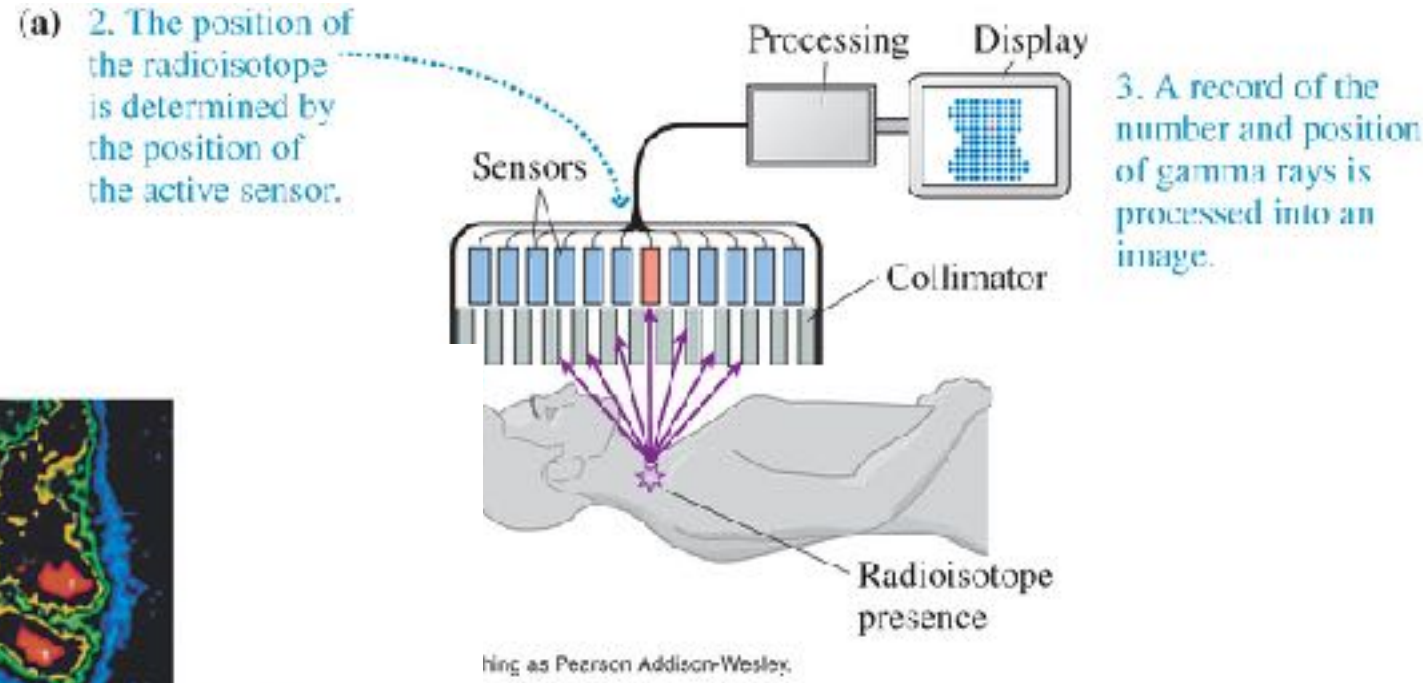
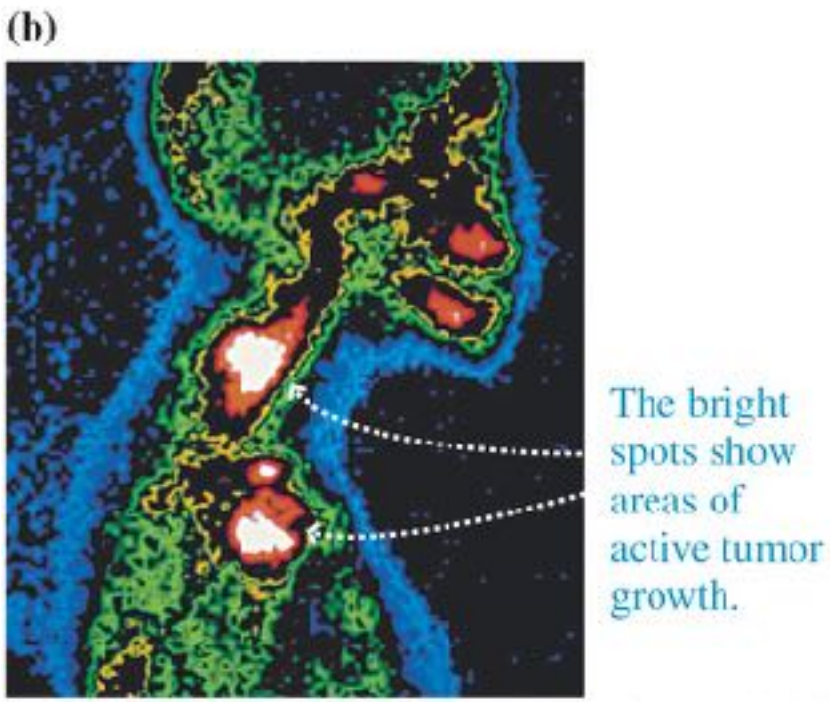
Nuclear Medicine

Locating the origin of a gamma ray requires segmenting and collimating detectors





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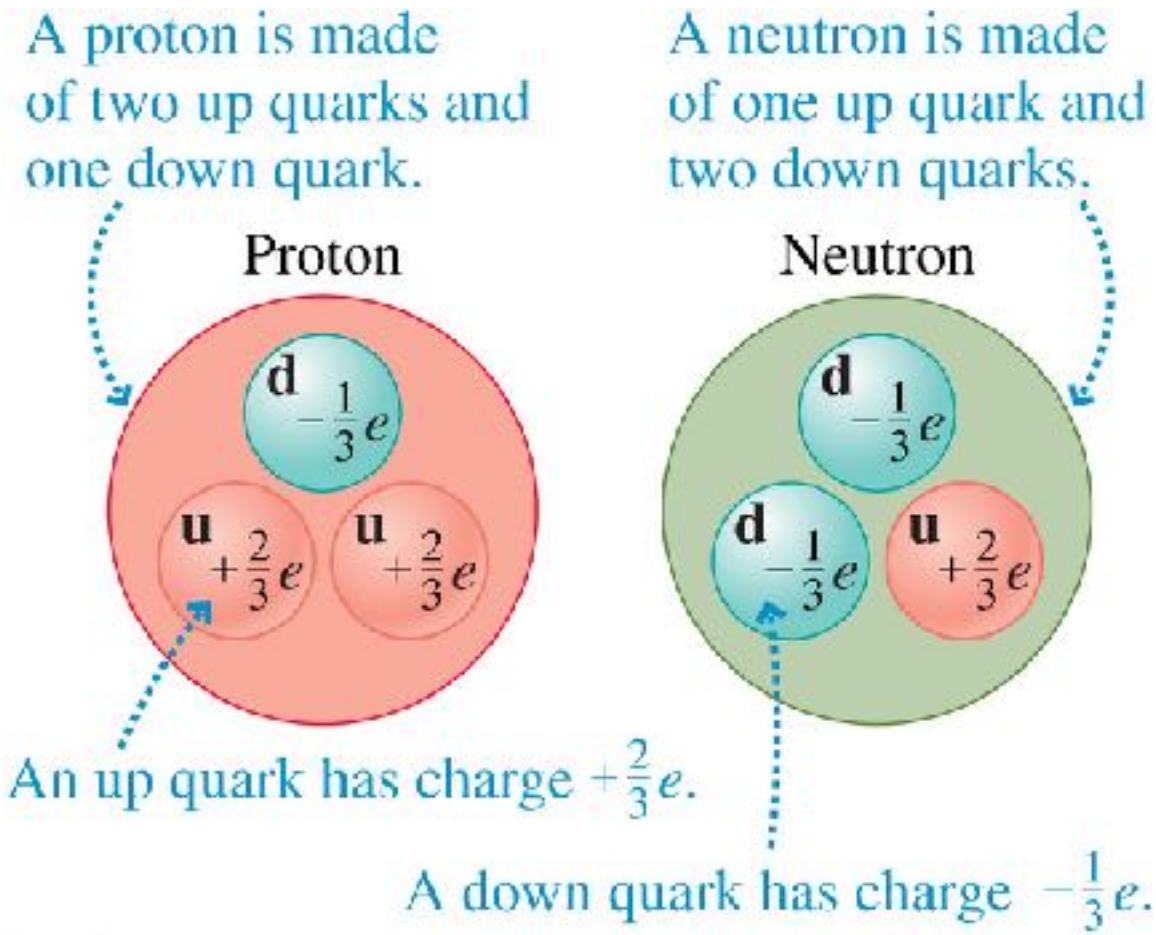
In the example here, the ^{99}Tc is enhanced showing the presence of an early stage- tumour.

Particle Physics and non-invasive imaging

Protons and neutrons have further substructure



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Beta decay is either

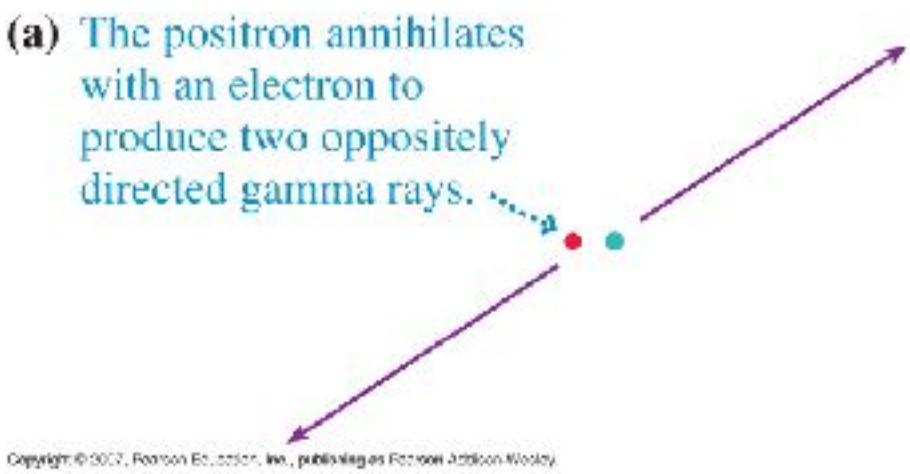
(electron emission)

or

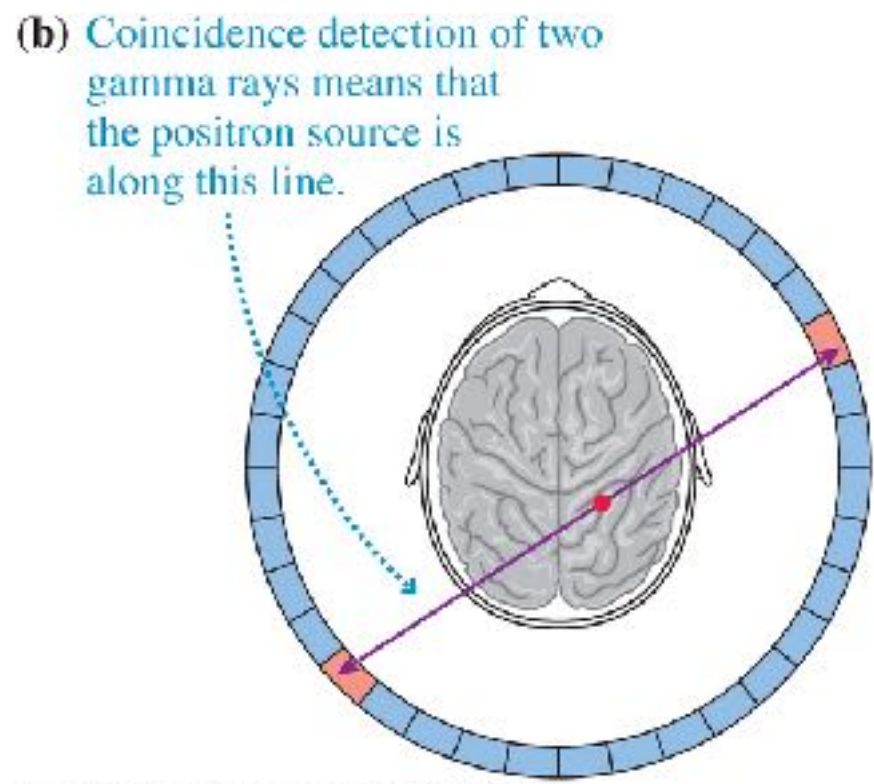
(positron emission)

Positron Emission Tomography

When a positron collides with an electron both are annihilated and two Gamma rays are emitted



These back-to-back gamma-rays can be used to locate the place in the body where the annihilation occurred

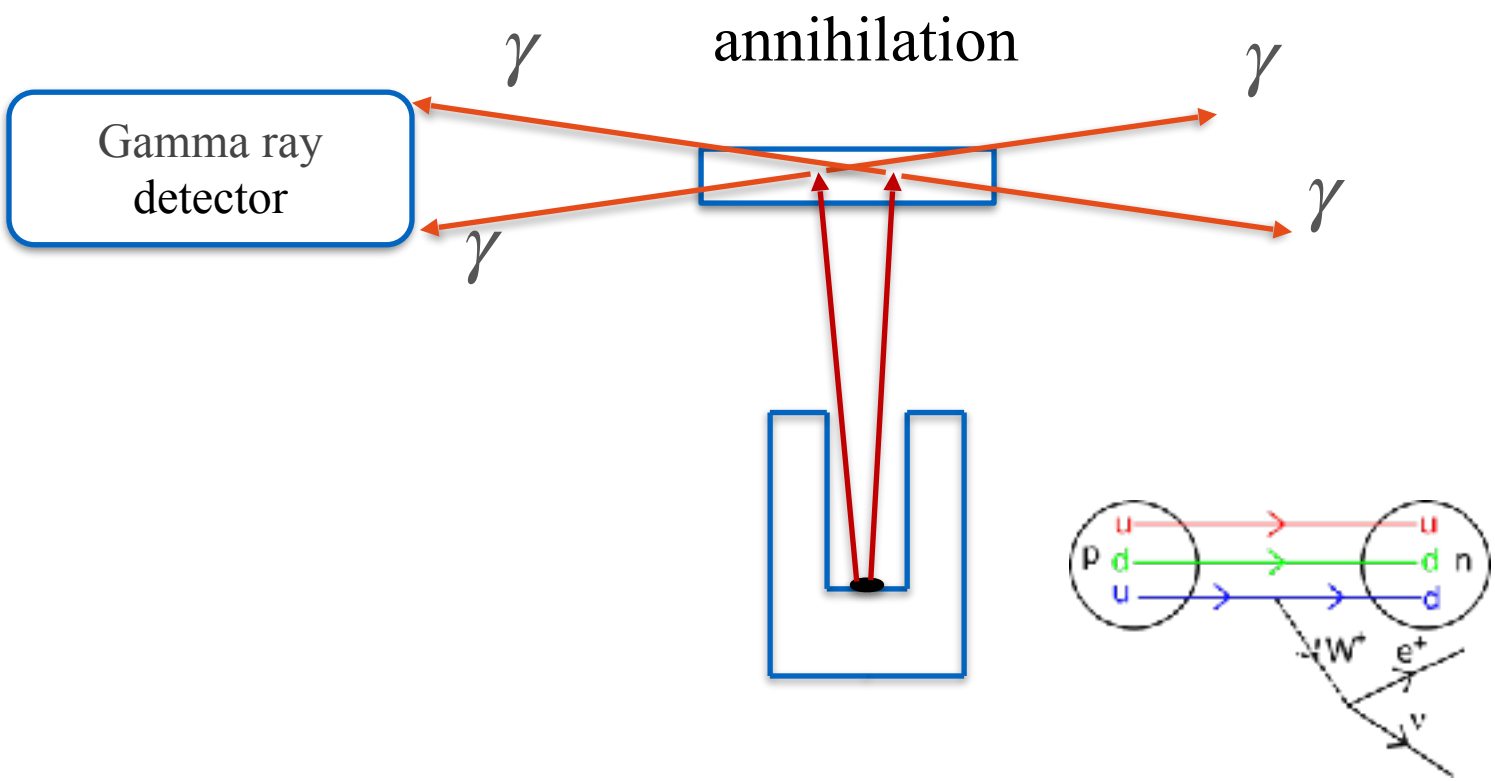


Tag a biologically active molecule
Eg Glucose with ^{18}F (positron emitter)

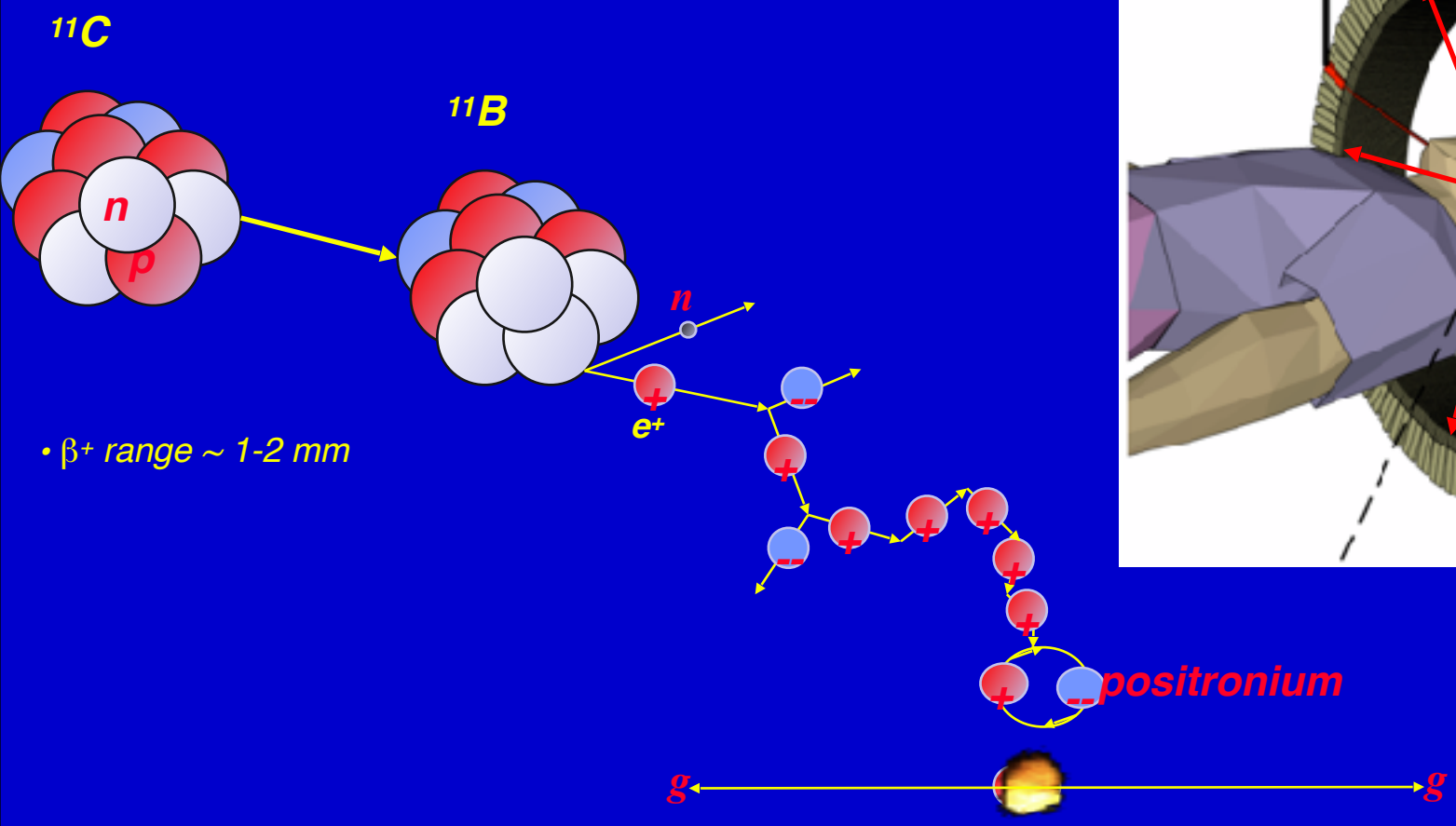
Or provide a sample of ^{15}O (positron emitter) and determine where in the body these molecules are absorbed.

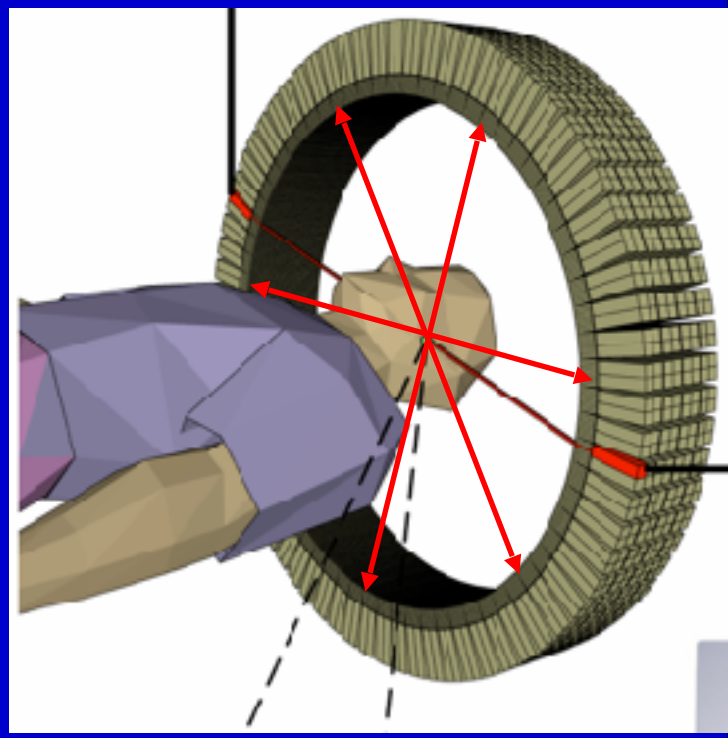
Can use this information to determine the functional activity of the body

Experimental Test of matter-antimatter annihilation



Applied Particle Physics!



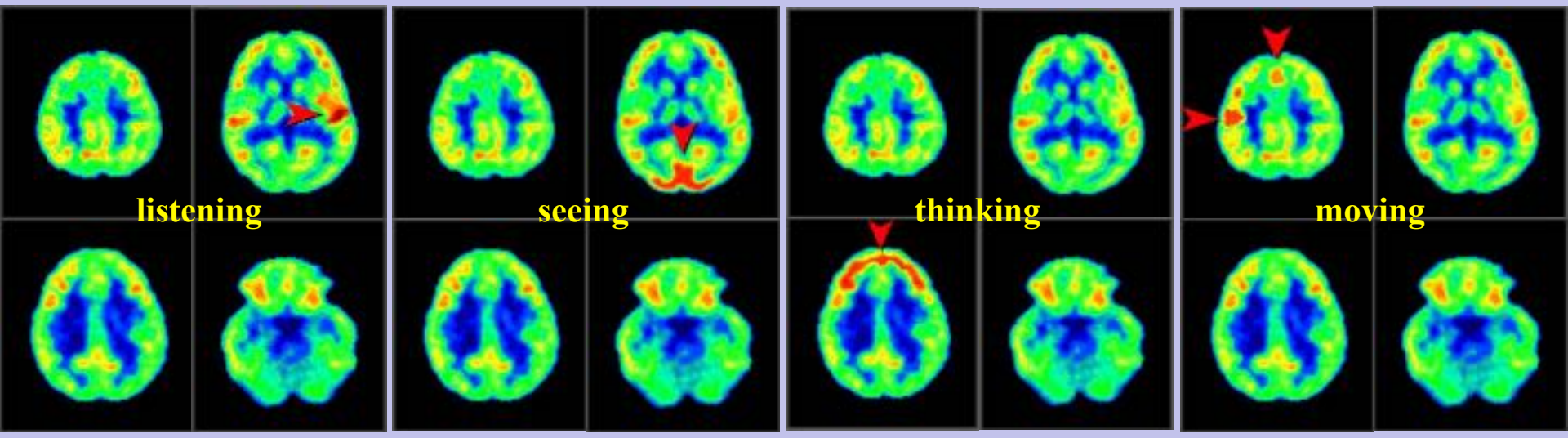


- β^+ range ~ 1-2 mm

Austin Health

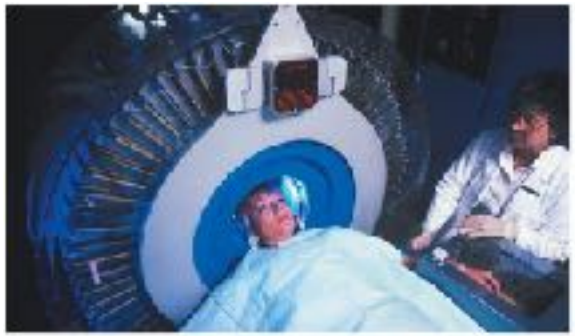
Observing Brain Activity using PET

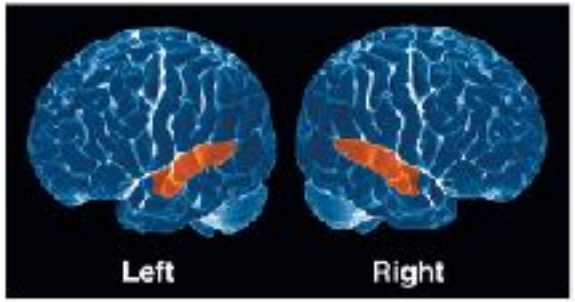
Tag FDG (bioactive chemical) with ^{18}F

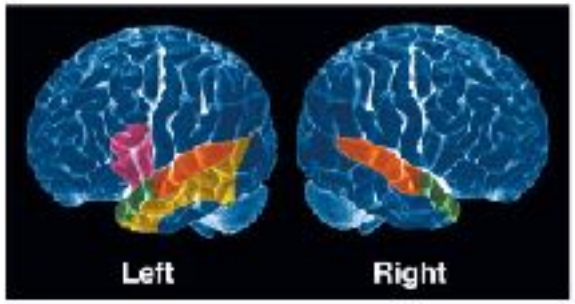


Austin Health

PET scans as a diagnostic tool







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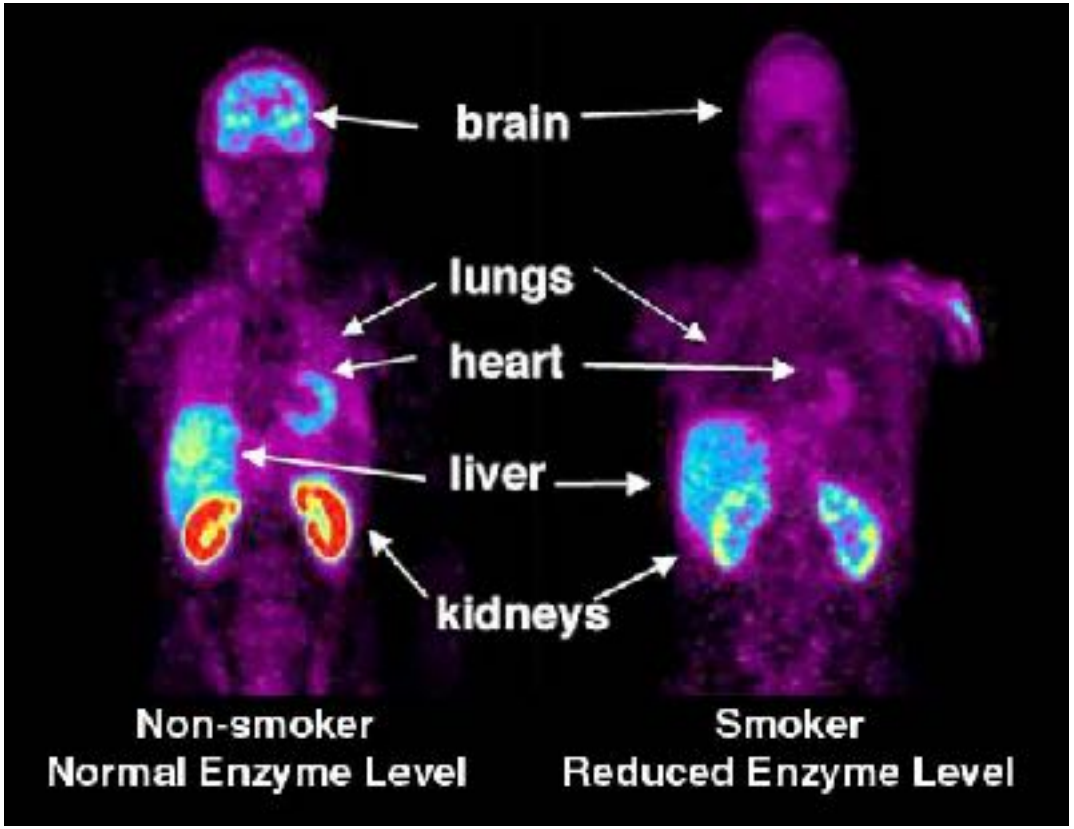


Ultrasound Imaging

All the techniques described previously use Ionising Radiation to probe bodily structure

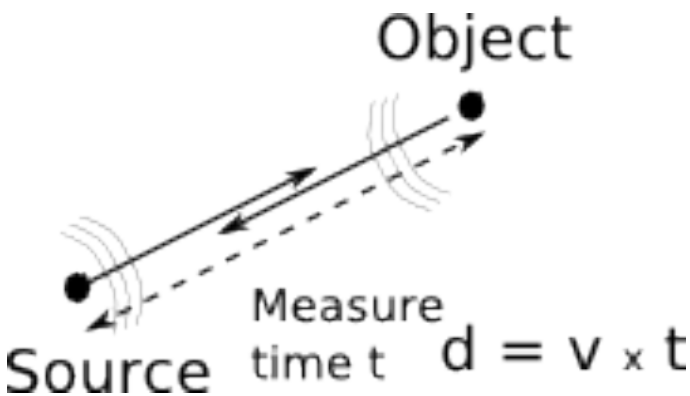
This is not suitable for investigating a developing foetus because of their extreme sensitivity to radiation.

Instead ultrasound imaging is used.

Here high frequency sound waves are emitted from a transducer placed external to the body.

These are directed through the body cavity under investigation.

The imagining works by detecting the strength and time of the reflections From internal structures.



With the distance and the angle of emission an image of the cavity is obtained

Ultrasound Imaging

High frequency is used to provide good spatial and temporal resolution

Exploits the reflection property of waves to build an image.

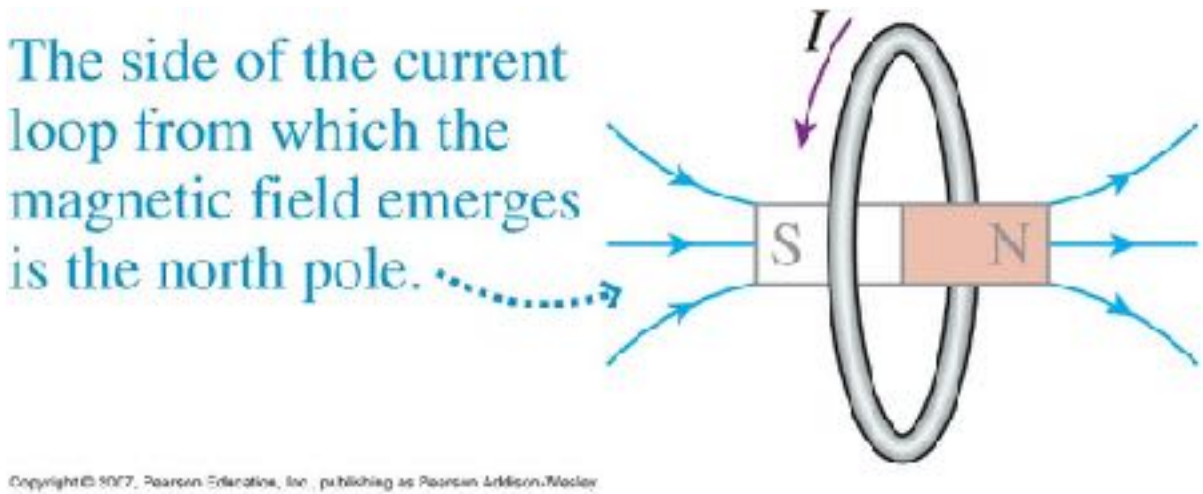
Some 2-D ultrasound images

3-D Ultrasound imaging

3-D imaging employs transducers that emit into a cone instead of a plane.

Magnetic Dipoles

The simplest source of magnetic field is loop of current.



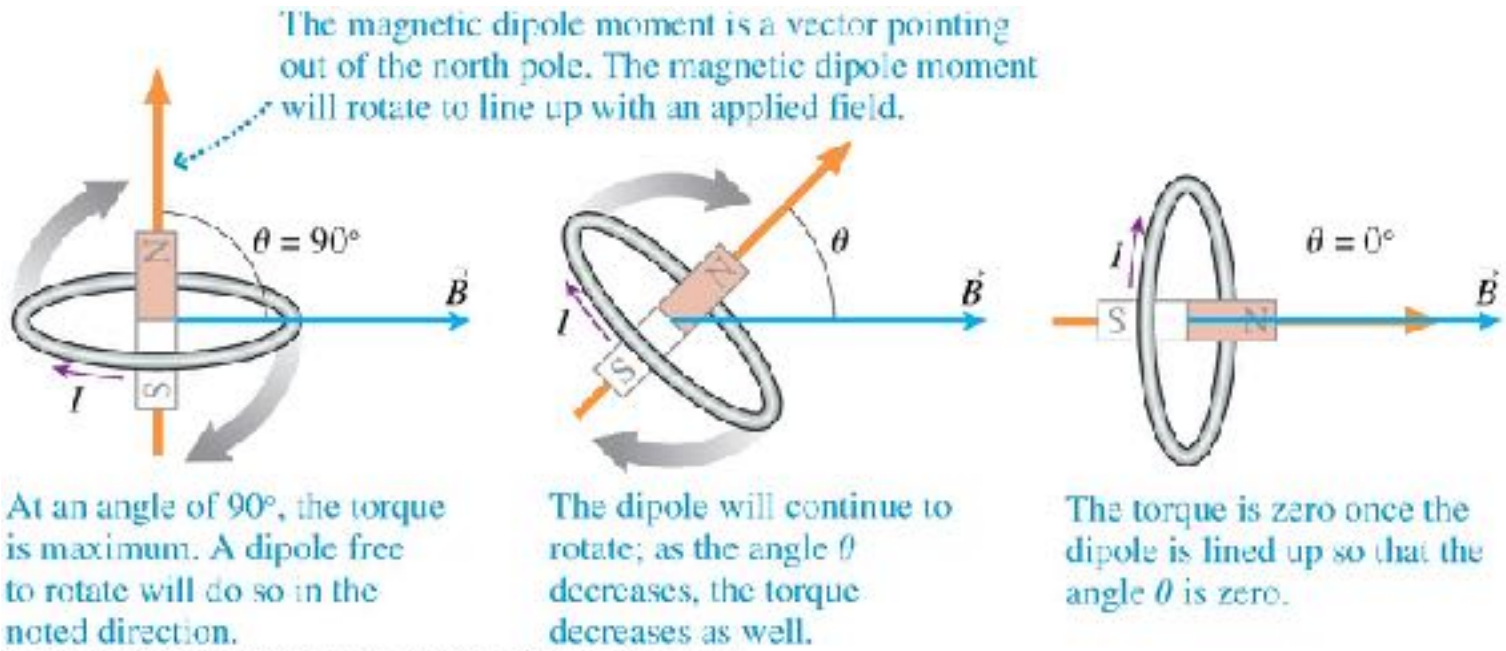
The strength of the magnetic field is characterised by the product of the Current in the loop and the area enclosed by the loop. This quantity is called the *Magnetic Dipole Moment*, μ.

Magnetic Dipole Moment $\mu = I A$

Where I = current in the loop
A = Area of the loop.

Magnetic Dipoles in an External Field

Like compass needles, magnetic dipoles twist to align with an external magnetic field



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If dipole is aligned with the magnetic field it is in a lower energy state than if it is pointing against the magnetic field.

The Energy difference, ΔE, between the two states is given by:

$$\Delta E = \mu . B$$

Where m = magnetic dipole moment
B = External Magnetic field

Protons are magnets!

Protons have a magnetic moment of: $\mu_p = 14.106067 \times 10^{-27}$ Amp.m^2 or J/T

If the proton is placed in an external magnetic field of 1 Tesla, the difference in energy is:

$\Delta E = 2 \times 14.1 \times 10^{-27} \times 1\ T = 28.2 \times 10^{-27}$ Joules

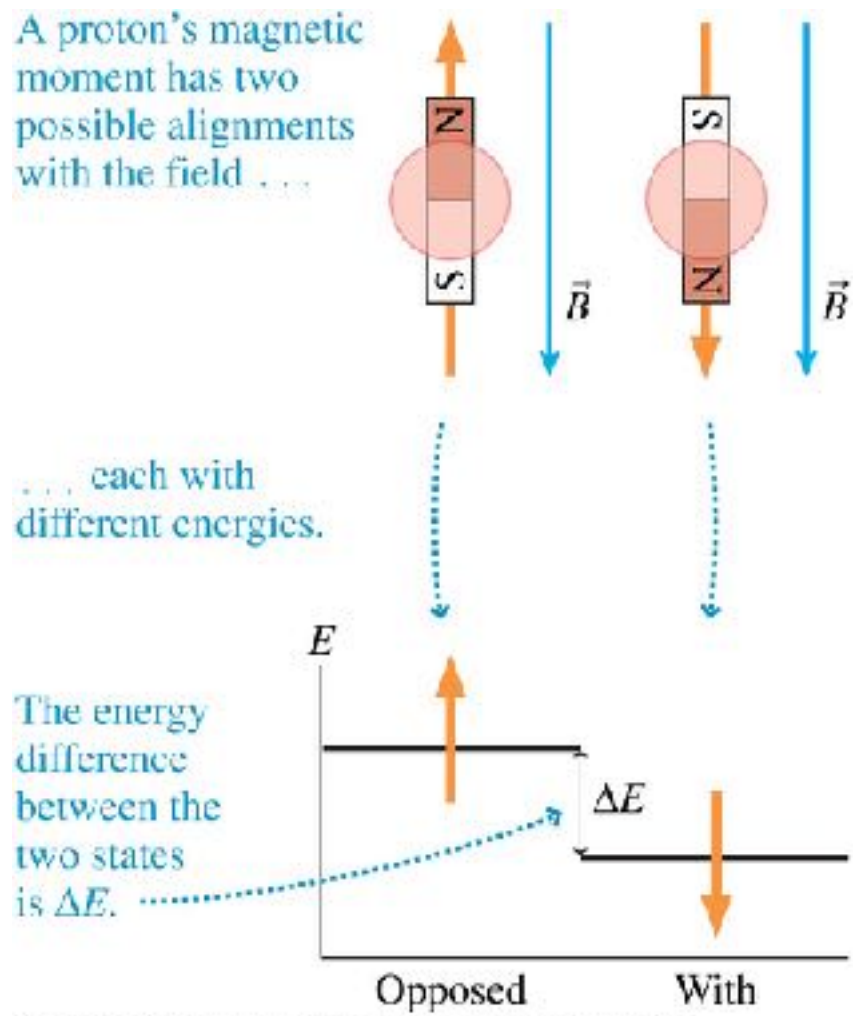
What frequency of E/M radiation is this?

Use E = hf

$\rightarrow\ 28.2 \times 10^{-27} = (6.63 \times 10^{-34}\ J.s)\ f$

$\rightarrow\ f = 28.2 \times 10^{-27} / 6.63 \times 10^{-34} = 42.56$ MHz

This frequency is comfortably in the Radio Wave range.



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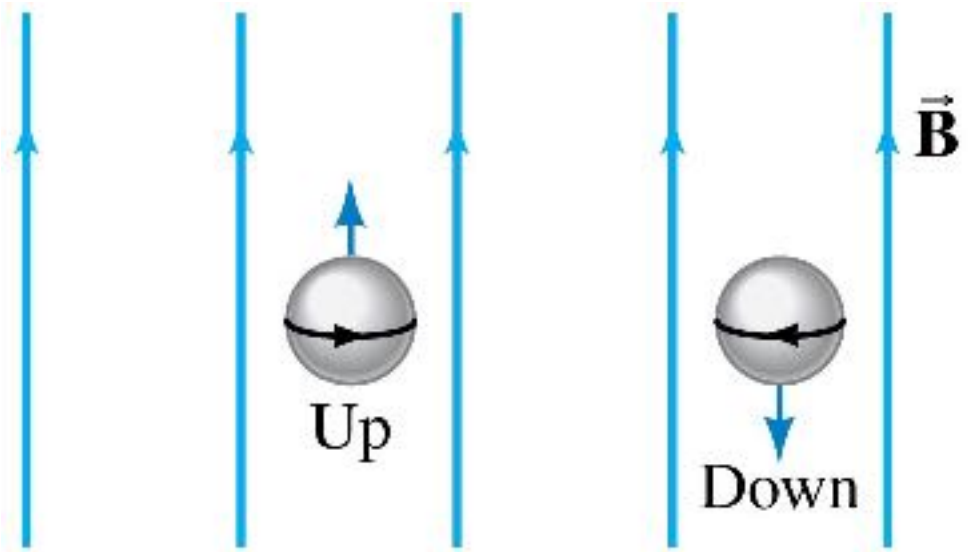
Nuclear Magnetic Resonance Imaging - G31.9

A proton in a magnetic field can have its spin either parallel or antiparallel to the field.

The field splits the energy levels slightly; the energy difference is proportional to the field magnitude.



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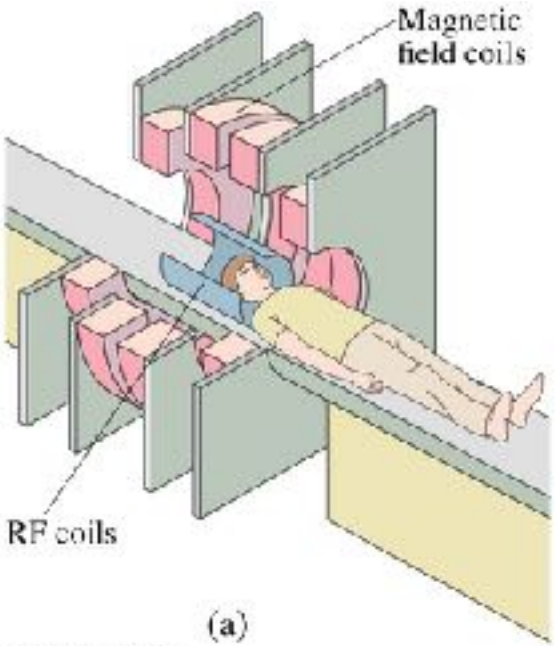
Na-5 Nuclear Magnetic Resonance

Nuclear Magnetic Resonance Imaging (MRI)



(a)

(b)

Patient is placed in a strong Magnetic Field (~1 Tesla)

The area is bathed in radio waves near the energy difference between protons aligned and opposite the magnetic field

The magnetic field is adjusted to make a small region precisely match this energy difference.

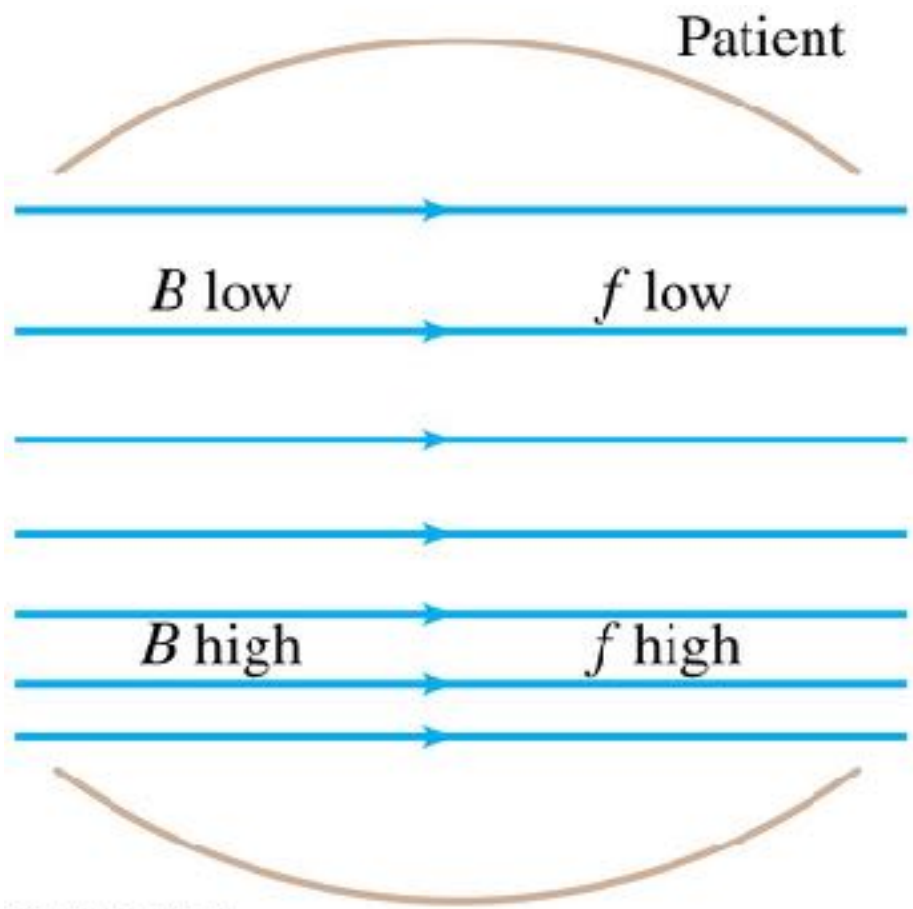
Protons in this region flip to point opposite the magnetic field

They then fall-back to become aligned, emitting radio-waves as they do

These radio waves are picked up by a receiver. The strength of the signal received gives the proton density at that point in space.

Nuclear Magnetic Resonance Imaging (MRI)

Giving the field a gradient can contribute to image accuracy, as it allows for the determination of the origin of a particular signal.

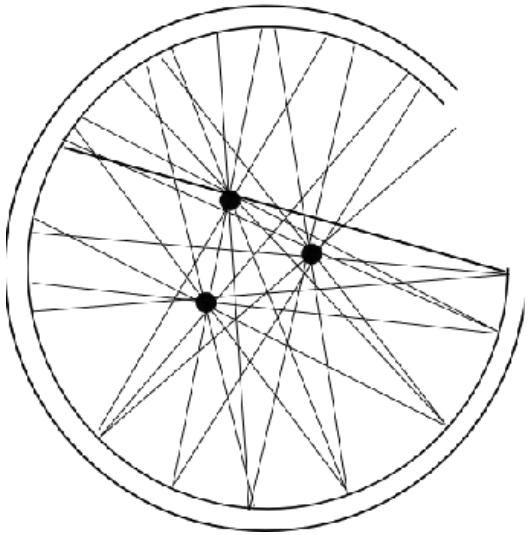


Summary

Contrast X-rays

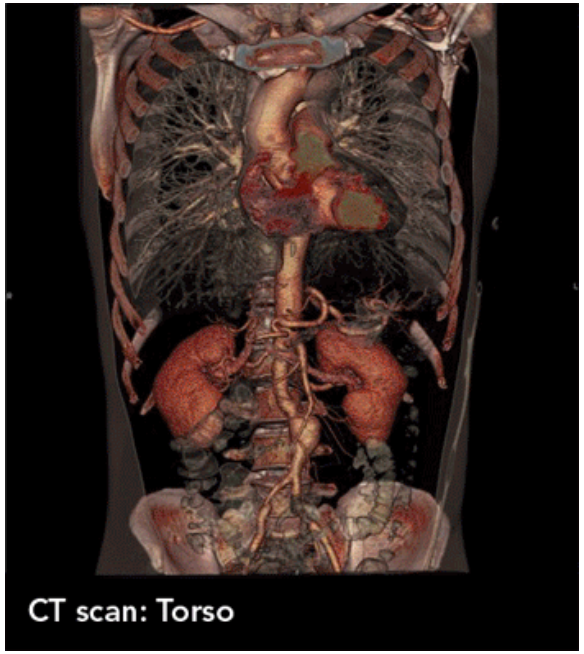
The probability of X-ray absorption increase as Z^2 where Z is the atomic number of the material

Computerised X-ray Tomography (CT-scans)



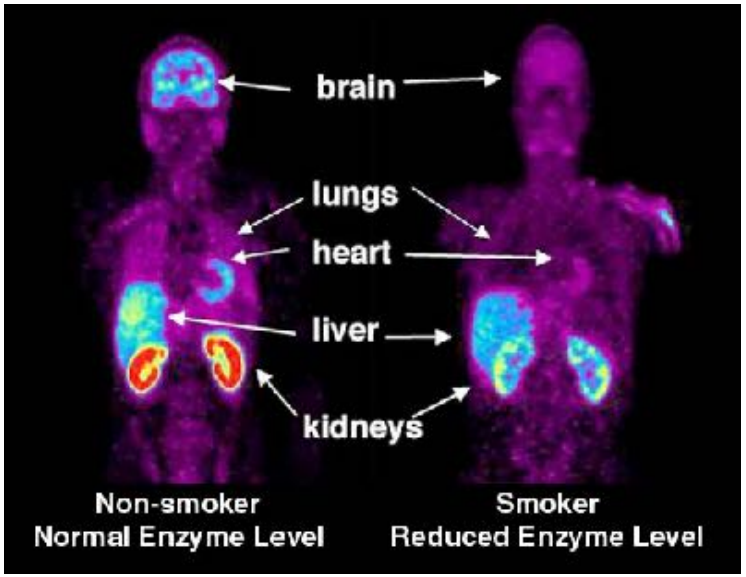
The intersection of response lines show the locations of Increased density

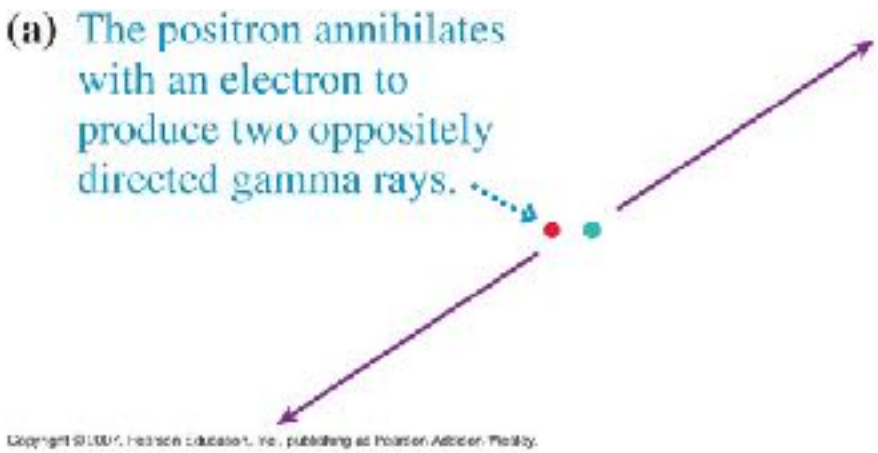
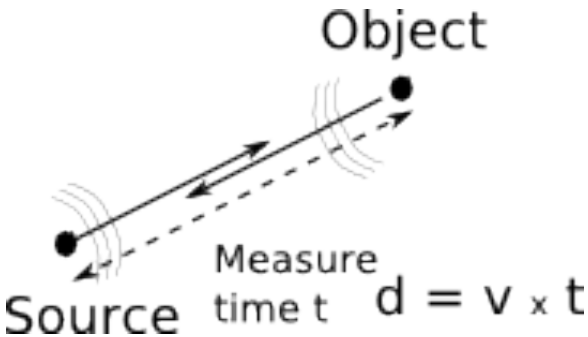
Contrast CT-Scans



Summary continued...

Positron Emission Tomography (PET-scans)





Ultrasound Imaging